

<< WORKFLOW >>

TEXTURE BAKING A 3D MODEL IN BLENDER

WHO IT'S FOR ...

X-Plane can't read Blender's proprietary paint schemes. You're a novice and want to convert your paint job – 'bake the textures' - of your painted 3D model, so XP can apply them. You need a primer, something to get you started.

ENHANCEMENTS

Feel free to suggest enhancements, improvements or corrections. Contribute your own workflows. Any learning is work in progress, the path from known > unknown. You are **freely encouraged to copy** this basic template/logo to produce and share your own workflows which could be useful to other XP developers, novice or otherwise. Just keep it simple, step by step, **so a 9-year-old can follow it**. Be **concise** (max info in minimum words). Test it. Include links. Screenshots really help.

It has an amber sidebar indicating 'intermediate' level difficulty, leaving green/red for beginner/advanced level practitioner workflows.

WHAT IT IS ...

This workflow lays out a basic approach for **baking the textures of a Blender 3D model to a single, exportable png file set**, which XP can read and apply to your obj8 file. It's designed for the non-expert, as easy as knitting socks, just to get you started. It assumes you have completed the design AND performed the UV Unwrap AND painted your model. It worked for me and I hope for you. Assumes Windows 10, but the principles should apply to other platforms.

REQUIREMENTS

- ✓ Blender 4.2.5 (or latest version compatible with the current Blender2XPlane add-on)
- ✓ the [SimpleBake](#) Blender add-on (purchase, then simply drag the zip onto Blender).
- ✓ your 3D model, fully textured and painted
- ✓ a single, consolidated, non-overlapping UV map (see previous workflow sheet)

OBJECTIVE

Our goal is to **'bake' a set (albedo/diffuse map, normal map, roughness map, metalness map) of png textures from our painted 3D model, to the UV map we created in the previous stage. This will convert Blender's proprietary texturing to the format required for X-Plane.** Rather than bake 'long-hand', manually, using Blender's complex procedural node system, we'll deploy the SimpleBake add-on, which simplifies the process. Inexpensive and worth it. It will provide the novice with the first rung on the ladder for texture baking.

RESOURCES

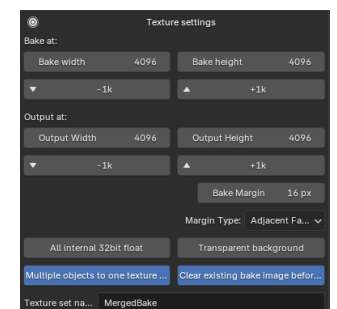
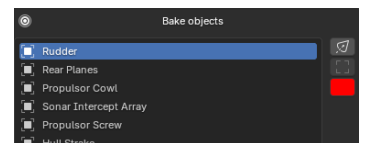
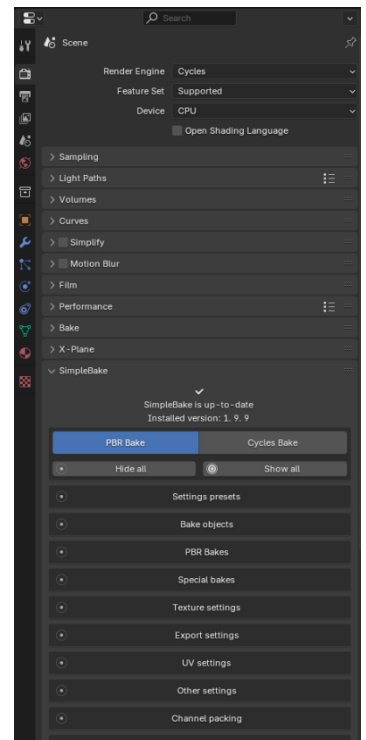
- ▶ [SimpleBake Overview](#)
- ▶ [Overview of SimpleBake Commands](#)

Green = Keystrokes

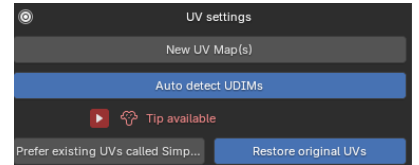
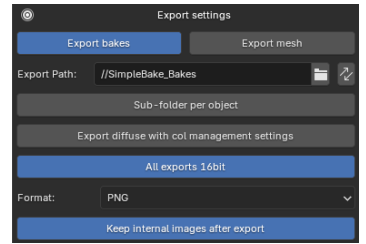
Blue = Menus/Commands

METHOD

- ▶ open your model in the main **3D Viewport** window
- ▶ Save a pristine copy of your .blend file as a fallback. Create a copy, name it - let's work on that. Backing up is good practice and will protect your work if everything goes belly up, rather like the climber knocking in a piton to protect from falling.
 - work through your objects/meshes in the **outliner** window and apply any/all modifiers you have used
- ▶ we'll be working in the '**Render Properties window**', so click on the  tab at the lower RHS of Blender's properties area. Note the SimpleBake sub-menu has been installed and is up to date. [It can be updated in Blender's main menu: **Edit / Preferences / SimpleBake**]
 - select 'Cycles' as your Render Engine
- ▶ if you follow the standard Blender approach to baking, it requires a myriad of procedural nodes, creating a 'spaghetti' of tangled and complex interconnected boxes with multiple settings. SimpleBake is going to do all this for us without the headache.
- ▶ Let's set up the bake by entering the following parameters / settings:
 - select '**PBR Bake**'
 - in your model's hierarchy window (above the properties window), select/highlight all objects/meshes you want to bake (**left click** the first, **Shift** and **left click** the last)
 - open the '**Bake objects**' menu, click '**Add**'. These should appear in the dialogue box. Check them – is this what you wanted to bake?
 - open the '**PBR Bakes**' menu. Select what png's you want SimpleBake to create for you. Here, I've chosen Diffuse (the basic colour albedo map), Normal (to create artificial bumps), roughness and metalness.
 - open the '**Texture Settings**' menu.
 - choose your desired resolution eg 4096x4096. I set '**bake**' and '**output**' to the same size, but you can play with this.
 - transparent background? Your choice ...
 - '**Multiple objects to one texture**' will consolidate all object UV's into one png. I did this for Astute.
 - '**Clear existing bake**' removes any legacy bakes
 - Use '**Texture Set Name**' to name your output files



- open the 'Export Settings' menu.
 - 'Export Path' - choose the directory you want to save to
 - Format 'PNG' as file type
 - 'Keep Internal Images' – I think this preserves your originals
- open the 'UV Settings' menu'
 - Select 'Restore original UV's'



Channel Packing Menu

- XP takes roughness and metalness settings from information 'packed' into the alpha and blue channels of the object's normal map.
- However, Simplebake packs to the diffuse png, *not* the normal map. Fortunately however, there's a solution ... **MakePBR**, see below.

- ▶ SimpleBake is now calibrated to produce your png texture set. Let's bake it.
 - Take your mouse pointer to the top LH corner of the 3D viewport window.
 - use the **+** and left click and drag out a new window to the right.
 - In the top LH corner of this window, use the 'Editor Type' drop-down list to choose 'Shader Editor'. This where nodes are added, connections made. SimpleBake is about to do the complex stuff for you, but it needs this window open to work.
 - Go back to the SimpleBake menu on the RH side and click 'Bake'. Watch – amazed - as nodes are created in the Shader window. No, I didn't understand them either 😊
- ▶ Open your chosen file path location to view your baked png files.
 - These can (if necessary) be edited in your chosen image editor, such as Photoshop, GIMP. I needed to correct a fault on my normal map (caused by my poor topology on the 3D model), simply using the clone brush to correct them.
 - **MakePBR**, a [downloadable](#) app, takes your roughness and metalness png's and packs them into the normal map in one click, thus completing our bake
 - Your texture png's are now ready for export to X-Plane / WED.

